Exact rectangular contraction and economical equality networks for 3-SAT

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Abstract

We review the twenty-nine placed refinement packets and five newer incoming packets at ProveIt commit `8e76d515063c`, with the application restricted throughout to 3-SAT. We do not obtain a smaller logarithmic-order worst-case exponent. We do obtain stronger exact contraction bounds and a more economical translation to 3-SAT.

The main additional ingredient is an exact-arithmetic transfer of the published complex matrix bounds $\omega \le 9/4$ and $\alpha > 0.465$ to the integer and truncated coefficient rings required by the SAT contractor. A fixed number-field realization of each finite bilinear algorithm makes this transfer possible; floating-point complex arithmetic is unnecessary. Combining the resulting rectangular bound with exact support certificates, including a new fifteen-plan cyclic construction, gives cached normalized time coefficients $28.6327662765\ldots$ at root degree $s^{9/10}$ and $10.3367239855\ldots$ at root degree $s^{1/12}$, on the scale $s/\sqrt{\log_2 s}$. The best corresponding coefficients in the reviewed packets are $29.8116089453\ldots$ and $13.3796490662\ldots$.

For a 3-CNF with L normalized literal occurrences, growing-degree equality trees have $(1 + o(1))L$ wires and polylogarithmic maximum degree. Consequently the new low-degree theorem gives an exact counter, and hence a decider, in

$$\operatorname{poly}(n+m)\, 2^{(10.3367239856\ldots+o(1))L/\sqrt{\log_2 L}},$$

where the slightly rounded decimal denotes an upper value; the exact coefficient is specified in Section 6. Its space coefficient is $10.3366545686\ldots$ on the same scale. The safe eventual coefficient 10.337 is certified for both resources. This statement pays for the SAT encoding; it does not substitute the number of variables for the wire budget. The archive supplies full proofs of the new interfaces and schedules, integer support certificates, exact rational numerical checks, a tested equality-network compiler, and a review of the referenced Lean development. The full subexponential counter and the imported matrix algorithms are not implemented or newly formalized here.

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1 Results and their precise scope

Let n denote the number of declared variables, m the number of input clauses, and L the number of literal occurrences after elementary normalization. Clauses have at most three literals. Repeated literals, tautologies, duplicate clauses and unused variables are handled explicitly. In particular $L \le 3m$; no assumption of bounded occurrence is imposed on the original formula. Logarithms in displayed complexity coefficients are base two. Polynomial factors include the time to read the original input.

The following two normalized statements concern the same Boolean tensor interface used by the previous packets, stated in Section 3. They are mathematical refinements using the cited exact contraction and matrix theorems as dependencies. They do not rest on an unproved new structured-summation hypothesis.

Theorem 1.1 (Two certified cached bounds)**.** *For a normalized network with at most s wires and root maximum degree at most $\lceil s^{9/10} \rceil$, there is a deterministic exact contraction algorithm whose time and space are at most*

$$(s+\mathcal{L})^{O(1)} 2^{(C_{\mathrm{gen}}+o(1))s/\sqrt{\log_2 s}}, \qquad C_{\mathrm{gen}} = 28.6327662765359\ldots.$$

If the root maximum degree is at most $\lceil s^{1/12} \rceil$, the time and space coefficients respectively become

$$C_{\mathrm{low}} = 10.3367239855467\ldots, \qquad C_{\mathrm{low,space}} = 10.3366545686347\ldots.$$

The same original entry-evaluation premise is permitted in both statements. Safe eventual coefficients are 28.633 in the first case and 10.337 in the second. All coefficients are specified exactly by rational data, square roots, and $\kappa = \pi\sqrt{2}/\ln 2$ in Section 6; decimal strings are descriptive.

Theorem 1.2 (Occurrence-based 3-SAT bound)**.** *Every 3-CNF can be decided, and its satisfying assignments counted exactly, in time*

$$\mathrm{poly}(n+m)\, 2^{(C_{\mathrm{low}}+o(1))L/\sqrt{\log_2 L}},$$

with space coefficient $C_{\mathrm{low,space}}$ on the same scale, as $L \to \infty$. Bounded L and formulas without nonempty clauses are handled directly. In particular, for $m \to \infty$ the time is at most

$$\mathrm{poly}(n+m)\, 2^{(3C_{\mathrm{low}}+o(1))m/\sqrt{\log_2 m}}, \qquad 3C_{\mathrm{low}} = 31.0101719566402\ldots.$$

For fixed $D > 0$, $m \le Dn$ gives the conservative time bound $\mathrm{poly}(n+m)2^{(3DC_{\mathrm{low}}+o(1))n/\sqrt{\log_2 n}}$.

The count statement is used because the imported contraction identities compute a count. Its only decision output is whether that integer is positive. No additional SAT width or problem class is part of the application.

1.1 A fair comparison against the previous proposals

Same normalized premise	Previous time	New time
Root degree at most $s^{9/10}$, cached	29.8116089454	28.6327662766
Root degree at most $s^{1/12}$, cached	13.3796490663	10.3367239856

The preceding bounds are from [4, 5]. The decreases in these certified leading coefficients are about 3.95% and 22.74%. They are comparisons of proved upper-bound formulas under matched assumptions, not measurements of SAT runtime. The new cached algorithms use subexponential space. The polynomial-space coefficient $46.1848455867\ldots$ under the general premise is not improved by Theorem 1.1 under a polynomial-space guarantee.

The preceding low-degree polynomial-space contractor has coefficient $16.8122338018833\ldots$ [4]. Our normalization also gives its occurrence-based consequence with that coefficient per literal. This is a tighter SAT translation, not a new polynomial-space contractor.

To avoid overstating the gain, one should also grant the old contractors the best available elementary normalization. High-occurrence branching already appears in [9]; changing its cutoff to $\log^2 L$ also exposes the low-degree interface with only a lower-order branching cost. Thus a fair composed previous cached benchmark is $13.3796490663\ldots$ per literal, even though the latest explicit translation used trivalent equality trees and up to $3L$ wires. Our stronger $10.3367239856\ldots$ coefficient improves that composed benchmark as well. The equality-tree saving is not counted twice.

1.2 What remains unresolved

Combining the new contractor with the existing dense-formula reduction retains the worst-case order

$$\mathrm{poly}(n+m)\,2^{O(n\log\log n/\sqrt{\log n})}.$$

The earlier density-sensitive envelope

$$\mathrm{poly}(n+m)\,2^{O\left(\frac{n}{\sqrt{\log n}}\min\{\log(2+m/n),\log\log n\}\right)}$$

can still be used [10, 11]. A fixed portfolio of the direct and compressed algorithms preserves the better applicable bound. The numeric constants in Theorem 1.2 do not replace the unspecified constants in that dense reduction.

No smaller logarithmic-order bound, such as $2^{O(n\log\log n/\log n)}$, has been proved here. No dominance is claimed over the unspecified numerical constant in the independent Anthropic algorithm. The improvement is substantial within the quantified contractor frontier, but it is not the requested change of worst-case exponent order.

2 Repository review and the formal baseline

2.1 Pinned scope and recent changes

The review uses these two immutable ProveIt snapshots:

Opening: `8e76d515063c4b9fbd8743a846b457e3a01bc039`
Closing: `0cf10036e6ad732c27f13e6058f4acd9b3d78415`

The opening snapshot contains twenty-nine filed refinement packets, the separate Claude manuscript, and five newer incoming ZIP packets. At the closing snapshot, those same five packets have been filed, giving thirty-four refinement packets plus the Claude manuscript; incoming contains only its README. We inspected the scoped commit histories, inventories, report sources and relevant implementations. This is not a claim that every prior proof has been independently reconstructed or formalized.

Commit	Relevant change
`2b925d8800`	Initial placement of seven refinement packets and manuscripts.
`3e09608991`	Vendors the OpenAI preprint from upstream `7ef5c0cc5c`.
`6d3adc4506`	Places eight round-two packets.
`450252e9d8`	Places five round-three packets and restores two scripts.
`a7ae5ab55f`	Places nine round-four packets.
`7de9d09fec`	Links the newly public Anthropic Lean development.
`26bd83d4b3`	Adds the five newest incoming packets reviewed here.
`8e76d51506`	Opening snapshot, containing the five incoming arrivals.
`324f3d9b00`	Places those same five round-five packets; no mathematical change.
`0cf10036e6`	Closing snapshot; thirty-four packets filed and incoming clear.

The relevant October 27 UTC commits occurred on October 26 in Pacific time. The archive records full hashes and immutable URLs. The source preprint directory is an ordinary vendored directory, not an unresolved submodule.

The closing audit compared all 213 newly placed file blobs with the original ZIP bytes and found them identical. The nine omitted members are a hash ledger, one duplicated README snapshot and seven generated build byproducts. Every previously filed research directory retains its Git tree hash. The latest histories of both requested paths end at the round-five placement, so no additional mathematical packet or changed mathematical content entered the comparison. Full evidence is recorded in `provenance.json` and `audit/closing_archive_verification.json`.

2.2 The preceding frontier

The source's sharper tensor-contraction appendix gives, for 3-SAT, $\mathrm{poly}(n+m)2^{O(n \log\log n/\sqrt{\log n})}$ time, with polynomial space, and $2^{O_D(n/\sqrt{\log n})}$ at fixed density [1]. Its interpolation-only body and the older OpenAI Lean scope statement are weaker; they are not the baseline used here.

The following ledger records the contribution and limitation of the twenty-nine packets already filed at the opening snapshot. Constants are on the normalized wire scale unless stated. The status of the original papers remains that of externally supplied research; ProveIt's placement commits explicitly do not claim to verify their entire contents [2].

Packet	Contribution relevant to this review
`cauchy_mode_improvement`	Factorial integral Cauchy modes; smaller mode coefficient, unchanged order.
`eth_contraction_improvements`	Sharp Hermite counts and streamed packet extraction; polynomial-space coefficient 648.44.
`3sat_precision_refinement`	Sharp recovery loss; faster structured mode summation remains an unproved interface.
`sat_density_improvement`	Density-sensitive $n\log(2+m/n)/\sqrt{\log n}$ envelope.
`eth_density_refinement`	Independent overlapping density improvement; no unrestricted order change.
`3sat_refinement`	Scalar parity lift and smaller incidence-degree cap; compact second contraction remains conditional.
`3sat_asymptotic_analysis`	Restricted precision, enumeration and degree barriers; no better unrestricted exponent.

Packet	Contribution relevant to this review
`eth_asymmetric_packet_balancing`	Nearly lossless asymmetric packing, additive precision buffer; coefficient 76.909.
`3sat_round2_refinement`	Matching-potential packing and two-stage coefficient 84.066; scoped rank tests.
`eth_balanced_packing`	Nearly balanced whole-vertex labeling; coefficient 94.748; direct-incidence SAT application.
`eth_high_moment`	High-moment derandomization and degree-sensitive grouping; coefficient 94.748.
`eth_matching_refinement`	Matching-potential version of the same 94.748 coefficient.
`eth_packet_refinement`	Two-stage packing and additive gaps; coefficient 95.664.
`eth_recursive_refinement`	Actual residue-class Cauchy structure and low-degree corollaries; coefficient 133.207.
`eth_joint_cauchy`	Full-orientation Cauchy modes and pair-weight entropy; coefficient 319.69.
`sat_adaptive_contraction`	Adaptive streamed separators; polynomial-space 55.709, cached 36.750.
`sat_frontier_improvements`	Variable alphabets and degree-sensitive schedules; polynomial-space 63.857.
`subexponential_sat_frontier`	Polynomial-space 63.852, cached 37.930; exact labeling and precision kernels.
`subsat_contraction_tree`	Fully charged cached tree construction; time 35.355.
`subexponential_sat_kernel_improvements`	Compatible compositions, adaptive Smith separation and exact extraction kernels.
`sat_adaptive_bounds`	Sharp flag separator; general polynomial-space 46.185 and low-degree cached 13.379649.
`sat_adaptive_tree_certificate`	Per-allocation tree choice; cached 30.891.
`sat_adaptive_frontier`	Adaptive quartet conditioning and joint cached selection; cached 31.252.
`sat_adaptive_matrix_partition`	Optimal specified sixteen-group separator; cached 34.325/34.024.
`subexponential_sat_adaptive_four`	Four-block local inequality; cached 32.916, or 32.856 with a sharper backend.
`subsat_rectangular`	Rectangular multiplication with input/output cost, ring bridge, and precision barriers; cached 33.994.
`eth_adaptive_exact_kernels`	Degree-sensitive schedules and exact packing/Cauchy kernels; general polynomial-space 55.332.
`3sat_round3_frontier`	Low-degree polynomial-space 23.666 and exact extraction kernels; reviewed an earlier frontier.
`eth_adaptive_schedules`	Variable schedules and incremental matching potential; polynomial-space 63.781.

Several packets report measured improvements of individual kernels. Those measurements are not end-to-end 3-SAT benchmarks and do not establish a new logarithmic-order theorem. Likewise, a stronger bound requiring an unsupplied factorization is not a proved competitor in the frontier table.

The five packets that were incoming at the opening snapshot, now filed as round five, must also enter the comparison [3]:

Round-five packet	Cached time	Cached space
`sat_affine_cauchy`	30.41335	29.46603
`sat_affine_contraction`	30.25704	25.98907
`sat_affine_geometry`	30.26987	29.48754
`sat_affine_plane`	30.69134	29.46603
`sat_pairwise_certificates`	29.81161	29.81161

All five retain the same normalized exponent scale. The last also improves the low-degree polynomial-space coefficient to $16.8122338\ldots$. The earlier 25.98907 cached space coefficient remains a useful time–space tradeoff: our generic 28.633 space guarantee does not dominate it. The principal numeric improvement claimed here is time under matched premises, together with a stronger low-degree time/space pair.

2.3 What the cited Lean development establishes

The supplied Anthropic SHA `1a6eee293b9e00cb432eef6018a21f523623a9af` is the exact PR 31 head. It was merged as `16a5ce41304fcdaea4e0be7c132e1c361dc26da5`. The formal statement quantifies one deterministic finite-alphabet machine and one constant before the formula and clause width. Specializing to 3-SAT gives exact counting within

$$2^{O\left(\min\left\{m/\sqrt{\log m},\ n\log\log n/\sqrt{\log n}\right\}\right)}.$$

The formal promises include distinct clause lists, every variable occurring, $n \geq 4$, and $m \geq 2$; elementary preprocessing and the separate naming frontend handle general encodings. This is a uniform machine theorem, not merely a different machine for each desired exponential rate [15, 16].

The exact SHA has a successful GitHub Actions verification run, including the build and the declared Comparator checks [18]. A static source scan found the two intended challenge placeholders and no additional occurrences of `sorry`, `axiom`, `admit`, or `unsafe`. Lean and Lake are unavailable in this execution environment, so this work did not perform a fresh kernel build. The audit examines the endpoint and its principal interfaces, not each of the roughly 1,050 proof modules.

The formal proof does not certify the numeric coefficients of any ProveIt refinement. Its intermediate machine simulation can cost a fixed power of an operation budget, which preserves the stated big-O order but changes the constant in the exponent. The explicit comments in `Solution.lean` explain this distinction. Our coefficients instead come from the direct bit-operation accounting in the stated contraction and matrix interfaces.

3 The exact contraction interface used as input

The graph is bipartite and Boolean: every edge carries one bit. A vertex entry is specified by a uniform procedure. With E the edge set, the contraction is

$$Z = \sum_{x\in\{0,1\}^E} \prod_{v} T_v(x|_{\delta(v)}).$$

There are at most s edges and polynomially many vertices. Entries belong to $(\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^\ell)$, with $1 \le P_0, \ell \le s$. The description length is $\mathcal{L}$, and an original entry costs $(s+\mathcal{L})^{O(1)}2^{O(s^{9/10})}$ time and polynomial space. The constants in these premises are fixed. Clause and equality tensors are much cheaper than this allowance.

The unchanged algebraic ingredients are the Cauchy deformation and Hermite recovery of [1], the sharp precision and near-unit packing refinements [6, 7], and the full-block integral Cauchy modes of [8]. They operate over

$$R_p = (\mathbb{Z}/2^p\mathbb{Z})[U,V]/(f(U), V^\ell),$$

where f is monic and irreducible modulo two, with degree polynomial in s. Its degree, ℓ, the precisions, and coefficient descriptions remain polynomially bounded. A tensor retains its creation precision: later calls use the same canonical representatives of its creation-time entries. Canonical lifting is not assumed to preserve ring operations.

3.1 The imported local identities and entropy estimate

At a call with M wires, fixed group alphabet r, taper u, packet exponent a, and retained valuation $i < u$, put $D = 2^{a-i}$. The cited lemmas give child size $(2/r + o(1))M$, working precision

$$p_t \le (6 + o(1))MD/u,$$

and at most $(2/r^2 + o(1))M/D$ mode families of each index on each directed pair. Their usable weights are $a + 1 - i + 4j - 2\,\mathrm{popcount}(j)$. For an exact weight allocation, let $\mathcal{I}_{gh}$ be the finite mode list on directed pair (g,h); empty lists are skipped. Set

$$h_{gh} = \log_2 |\mathcal{I}_{gh}|, \qquad \mathcal{K}(M,u) = \kappa\frac{M}{\sqrt{u}}, \qquad \kappa = \frac{\pi\sqrt{2}}{\ln 2}.$$

The contribution is the exact finite sum

$$\sum_{\xi_{gh}\in\mathcal{I}_{gh}} \prod_{g=1}^{r} Z_g((\xi_e)_{e\ni g}) \pmod{2^{p_t}}. \tag{1}$$

Every Z_g entry is an actual recursive child contraction. It is not a free table lookup until its entry has been computed and stored.

For every vector $z \ge 0$ in a fixed finite support family, the imported common-tilt entropy bound is

$$\sum_{g\ne h} z_{gh}h_{gh} \le \left(\frac{\|z\|_2}{r} + o(1)\right)\mathcal{K}(M,u). \tag{2}$$

For unordered edges set $g_{\{a,b\}} = h_{ab} + h_{ba}$. Thus

$$v \cdot g \le \left(\frac{\sqrt{2}\|v\|_2}{r} + o(1)\right)\mathcal{K}(M,u). \tag{3}$$

The factor $\sqrt{2}$ must be retained. The generating-product logarithm behind this estimate has leading term $\pi^2/(24\theta)$; minimizing the precision term plus its common-tilt contribution gives (2). The additive precision buffer used below makes the error uniform over the finite support family and all active depths.

There are polynomially many weight allocations, since r is fixed. The list cardinalities, iteration, and ranking are obtained by the existing finite knapsack procedure, not by an additional counting oracle.

3.2 Packing and the fully charged recurrence

Write $L_p = 2^a$, $Q = r^a$, and let R, C be the two side-degree bounds. The sufficient packing conditions from [7] are

$$QR/M = o(1), \qquad Q^2C(\log s)^2/M = o(1), \qquad L_pQ^2/M = o(1). \tag{4}$$

The notation L_p distinguishes packet length from the SAT literal count. At the root both degree bounds are original. Below it the fresh side has degree at most $2^{a_{d-1}+1}$, and the retained side contains original tensors and earlier packets; no retained vertex gains ports.

Caching one singleton group table requires at most $2^{(\alpha_r+o(1))\mathcal{K}}$ paid recursive evaluations, where

$$\alpha_r = \frac{\sqrt{2(r-1)}}{r}.$$

If local table combination has coefficient β_r, the recurrence is

$$T_d \le \mathrm{poly}(s)\left(2^{(\alpha_{r_d}+o(1))\mathcal{K}_d}T_{d+1} + 2^{(\beta_{r_d}+o(1))\mathcal{K}_d}\right). \tag{5}$$

The maximum over all routes in this recurrence determines the global coefficient. In particular, improving only a local matrix bound need not improve the complete algorithm. We explicitly compute both the local routes and the terminal route in Section 6.

4 An exact rectangular backend from the new characteristic-zero bounds

The matrix-multiplication input is updated at this pinned OpenAI Math commit:

`7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a`.

The October 2 preprint [13] *An Upper Bound of* $9/4$ *for the Matrix Multiplication Exponent* proves

$$\omega_{\mathbb{C}} \le \frac{9}{4}.$$

Its proof bounds the exact-rank exponent, then selects one fixed finite exact rank scheme for every positive accuracy. Its final remark explicitly places the coefficients in a number field by the Nullstellensatz. The September 24 preprint [14] *Complex Matrix Multiplication Below* 2.258 *and Rectangular Bounds* proves

$$\alpha_{\mathbb{C}} > 0.465 = \frac{93}{200},$$

and its characteristic-zero dual-bound corollary transfers this statement to every characteristic-zero field. We use these two primary results as mathematical dependencies; their entire tensor constructions are not reproved here. The additional argument below supplies the exact bit-complexity and residue-ring interface that the arithmetic statements alone do not provide.

Put $\rho = 93/200$. Fix a rational accuracy $\varepsilon > 0$, independent of the SAT instance. In the numerical certificates we take $\varepsilon = 10^{-5}$. Every constant in this section may depend on this one fixed accuracy and on the resulting finite schemes, but not on matrix dimensions, entry bit lengths, the SAT input, or a recursion depth.

4.1 A finite exact scheme is effective

Lemma 4.1 (Effective algebraic schemes)**.** *For every rational $\delta > 0$, one can deterministically find a number field K and two fixed exact bilinear schemes over K as follows:*

1. *a square scheme for $u \times u$ multiplication, with $u \geq 2$ and rank budget $r_{\mathrm{sq}} < u^{9/4+\delta}$;*
2. *a rectangular scheme for the dimensions (a, b, a), with $a \geq 2$, $1 \leq b \leq a$, $b^{200} \geq a^{93}$, and rank budget $r_{\mathrm{fl}} < a^{2+\delta}$.*

The search is a finite preprocessing computation depending only on δ. No assertion of a practical search time or small scheme is made.

Proof. For the square scheme, the exact-rank exponent in the October 2 theorem is at most $9/4$, so some fixed u and exact rank budget satisfy the strict displayed inequality.

For the rectangular scheme, choose $k > \rho$ with $\omega_{\mathbb{C}}(1, k, 1) = 2$. Such a k exists by the strict dual bound. The exact-rank limit for rectangular matrix multiplication, proved in the September 24 paper, supplies arbitrarily large fixed values t and exact schemes of dimensions

$$a = \lceil e^t \rceil, \qquad b = \lceil e^{kt} \rceil, \qquad \log r_{\mathrm{fl}} < (2 + \delta/2)t.$$

One may take $k < 1$, by padding if necessary. For sufficiently large t, $b \leq a$ and $b^{200} \geq a^{93}$, since $k > 93/200$. Also $r_{\mathrm{fl}} < a^{2+\delta}$. Thus a finite scheme with the required integer inequalities exists.

For either specified dimension triple and rank budget, express each rank-one term as a product of three linear forms. Equating the coefficients with those of the matrix-multiplication tensor gives finitely many polynomial equations over $\mathbb{Q}$:

$$\sum_{\nu=1}^{r} A_{\nu,i} B_{\nu,j} C_{\nu,k} = T_{ijk} \in \{0, 1\}.$$

A complex solution makes this ideal proper. The Nullstellensatz gives a solution over $\overline{\mathbb{Q}}$, whose finitely many coefficients lie in a number field. Taking the compositum puts both schemes in one number field K.

This existence assertion is effective in the elementary, unbounded-search sense needed here. Enumerate finite dimension triples, rank budgets, number fields specified by irreducible rational polynomials, and rational coordinate descriptions of all scheme coefficients. Test the coefficient identities exactly in the proposed field and test the rank/shape inequalities by integer powers, since δ and ρ are rational. Enumeration by increasing finite description length visits every finite candidate. Exact field arithmetic and irreducibility testing over $\mathbb{Q}$ are effective. An algebraic solution described above will eventually be visited, so the search terminates. One can instead use effective polynomial-system elimination, but no bound for that elimination is needed. After this preprocessing, the two schemes are fixed finite data in a single algorithm for all input sizes. □

4.2 Coefficient growth in the whole recursive algorithm

Lemma 4.2 (Bit cost of a fixed algebraic bilinear recursion)**.** *Fix finitely many exact bilinear matrix schemes with coefficients in a number field K, and use them by recursive blocking, tensor composition, permutation of the three tensor legs, zero padding, and ordinary blocking. Suppose*

every recursion path has length h, and let N be the largest matrix dimension. For integer input entries of at most B bits, all intermediate number-field coordinates can be represented exactly by rational numbers with $O(B + h + \log N + 1)$ bits in numerator and denominator. Consequently each scalar operation of the whole algebraic computation costs $(B + h + \log N + 1)^{O(1)}$ bit operations. Simulating in a fixed rational basis changes the whole arithmetic operation count by a constant factor; it does not multiply the rank budget by the field degree at every recursion level.

Proof. Choose a rational basis $e_1 = 1, e_2, \dots, e_d$ of K. The degree d is fixed. Write every scheme coefficient in this basis and write the fixed structure constants in

$$e_i e_j = \sum_{k=1}^{d} \mu_{ijk} e_k.$$

Choose an integer $D \geq 2$ clearing their finitely many rational denominators. Then

$$\mathcal{A} = \bigoplus_{i=1}^{d} \mathbb{Z}[1/D] e_i$$

is a subalgebra of K containing all coefficients. There is a fixed constant C bounding all relevant coordinate magnitudes, the number of terms in each fixed linear form, and the number of coordinates used in one product.

In a recursive bilinear scheme, the operands passed down any branch are linear combinations of the original left inputs and, separately, of the original right inputs. At one level each coordinate is a fixed linear combination of a bounded number of coordinates. After j levels its denominator divides D^{cj}, and its absolute value is at most $C^{cj} 2^B$, for a fixed c. Thus it has $O(B + j + 1)$ bits in rational coordinates. This conclusion follows equally by induction with a common denominator; reduced fractions are not required.

At a leaf, one multiplies a left linear form by a right linear form. The two denominator exponents add and the coordinate structure constants add only a fixed amount. The resulting coordinate numerators have $O(2B + h + 1)$ bits. On the upward path only fixed linear combinations of these bilinear values are taken. Each level increases the logarithmic magnitude and common-denominator exponent by at most a fixed constant. Hence the final bound is $O(B + h + \log N + 1)$ throughout.

This argument uses the bilinear structure of the recursion: no sequence of repeated squarings of an arbitrary computed value is present. Equivalently, each leaf product has degree one in each original operand, and every later operation is linear in such products. The exponentially large expanded rank decomposition need not be stored or constructed. Ordinary block summation adds at most $O(\log N)$ to the bit length, because there are polynomially many summands in N. The ordinary recursive implementation already performs the operations whose heights were bounded.

A number-field addition or multiplication uses a fixed number of rational coordinate operations, because d and its multiplication table are fixed. Elementary integer arithmetic and exact rational normalization on $O(B + h + \log N + 1)$ bits take polynomial time. This simulates each node of the already chosen recursion over K. It introduces a constant factor per executed operation, rather than replacing a rank-r recurrence by a rank-(dr) recurrence. □

In particular $h = O(1 + \log N)$ for the fixed schemes on matrices of maximum dimension N. A depth-first bilinear implementation forms one child product at a time, adds it into its parent output, and releases it. At each ancestor it retains only a fixed number of input/output-sized arrays. Their

dimensions decrease geometrically down a nontrivial recursion path. Its bit space is therefore a polynomial factor in $B + \log N + 1$ times the largest input/output array, with a harmless additional logarithmic factor if an implementation retains all active descriptors.

4.3 Integer outputs and residue rings, including characteristic two

Proposition 4.3 (Exact lifting to the rings used by the contractor)**.** *Let p, ℓ, t and the bit lengths of all ring data be polynomially bounded in a parameter s. Let $f \in \mathbb{Z}[U]$ be monic of degree t, with polynomially bounded coefficient bit lengths, and put*

$$R = (\mathbb{Z}/2^p\mathbb{Z})[U, V]/(f(U), V^\ell).$$

For matrices over R whose dimension logarithms are polynomially bounded in s, any backend furnished by Lemma 4.1 can be evaluated exactly with only a factor polynomial in s overhead over its matrix arithmetic count. The factor's exponent and all scheme data are fixed independently of s. No denominator is inverted modulo 2^p.

Proof. First consider integer matrices. Run the entire fixed recursive algorithm over the number field K, represented in the rational basis above. The bilinear identities hold exactly, so on rational inputs the output lies in the one-dimensional rational subspace $\mathbb{Q}e_1$ and equals the ordinary product. On integer inputs it is an integer. All other coordinates are exactly zero. Project to the coordinate of e_1 and perform its final exact rational division. Its denominator divides $D^{O(\log N)}$, whose bit length is $O(\log N)$; its numerator and denominator have the bit bounds of Lemma 4.2. An exact integer quotient is obtained, not a numerical approximation. The integer D may be even. Nothing has been reduced modulo 2^p during this rational computation, so even denominators cause no problem.

For entries in R, take the canonical polynomial representatives

$$\sum_{0 \le i < t,\, 0 \le j < \ell} c_{ij} U^i V^j, \qquad 0 \le c_{ij} < 2^p.$$

Each coordinate is an integer matrix. Multiply every required pair of coordinate matrices by the exact integer procedure, collect equal monomials, discard powers V^j with $j \ge \ell$, and reduce the U-degree using the monic polynomial f. Finally reduce integer coefficients modulo 2^p. There are at most $(t\ell)^2$ coordinate matrix products, and all remaining operations are polynomially many array additions and scalar integer operations per entry. Monicity makes polynomial reduction exact over the integers, without division by a nonunit. The coefficient growth of this finite reduction is polynomial in s, since both its degree and the coefficient bit lengths of f are polynomially bounded.

The natural map $\mathbb{Z}[U, V] \to R$ is a ring homomorphism, so this procedure gives the required product in R. We do not assert that canonical lifting from R to integer polynomials is itself a ring homomorphism. We only lift the given inputs, compute their polynomial product exactly, and then apply the quotient map. This also leaves the contractor's immutable creation-precision convention intact.

Here input bit lengths and $\log N$ are polynomial in s, so Lemma 4.2 and the polynomial number of coordinate products prove the claimed overhead. CRT is optional and unnecessary: the exact rational/integer route already handles characteristic two, including the case in which two divides every natural denominator of the chosen algebraic schemes. □

4.4 The resulting nine-support rectangular bound

Theorem 4.4 (Updated exact rectangular backend)**.** *Let $x \geq y \geq z \geq 0$ be the base-two logarithms of three positive integer matrix dimensions, in decreasing order. Under the bit-size conditions of Proposition 4.3, the product can be computed exactly in a polynomial factor times $2^{F_\varepsilon(x,y,z)}$ bit operations, where*

$$F_\varepsilon(x,y,z) = \max\left\{(1+\varepsilon/2)(x+y),\quad x+\left(\frac{335}{428}+\varepsilon\right)y+\frac{50}{107}z\right\}.$$

Its bit space is bounded by a polynomial factor times 2^{x+y}, and hence also by a polynomial factor times $2^{F_\varepsilon(x,y,z)}$. For $\varepsilon = 10^{-5}$, this is exactly the maximum of the nine linear supports obtained by permuting

$$(1+\varepsilon/2, 1+\varepsilon/2, 0), \qquad \left(1, \frac{335}{428}+\varepsilon, \frac{50}{107}\right).$$

Proof. Choose the two schemes with a smaller fixed accuracy, for instance $\delta = \varepsilon/4$, so that logarithmic recursion overhead can be absorbed into the stated ε slack. Their coefficient fields are combined once, as above. First block the largest dimension down to the scale of the middle dimension, paying $2^{x-y+O(1)}$ blocks.

Suppose $z \geq \rho y$. Define

$$t_0 = \frac{y-z}{1-\rho}, \qquad t_1 = \frac{z-\rho y}{1-\rho}.$$

These are nonnegative, satisfy $t_0 + t_1 = y$, and decompose the remaining shape into a flat shape $(t_0, \rho t_0, t_0)$ and a square shape of side exponent t_1. Tensor the corresponding fixed recursive schemes. Rounding their recursion depths upward changes their large dimensions by fixed factors. The flat scheme may have $b > a^\rho$; this only pads the smaller dimension and it stays no larger than the large dimensions, because $b \leq a$. Thus the largest padded array still has $2^{2y+O(1)}$ entries per large-dimension block.

For clarity, the rounded depths can be chosen without an oracle for logarithms. If the middle and small dimensions are n_y, n_z, choose the least nonnegative h_0, h_1 satisfying respectively

$$a^{107h_0} n_z^{200} \geq n_y^{200}, \qquad u^{107h_1} n_y^{93} \geq n_z^{200}.$$

These are exactly the upward roundings of $t_0/\log_2 a$ and $t_1/\log_2 u$. Fixed-degree integer powers and comparisons determine them in polynomial bit time. They guarantee the required dimensions; the two rounding errors are bounded constants.

The fixed-scheme cost, with its logarithmic recursion factors absorbed by the reserved accuracy, is at most

$$\begin{aligned} x - y + (2+\varepsilon)t_0 + (9/4+\varepsilon)t_1 &= x + \left(1 - \rho\frac{1/4}{1-\rho} + \varepsilon\right)y + \frac{1/4}{1-\rho}z \\ &= x + \left(\frac{335}{428}+\varepsilon\right)y + \frac{50}{107}z. \end{aligned}$$

Here an additive constant in the exponent is absorbed in a fixed multiplicative constant. If $z < \rho y$, pad to the flat shape. Its exponent is at most $x+y+\varepsilon y$, which is bounded by $(1+\varepsilon/2)(x+y)$ because $x \geq y$. These two cases prove the time bound. The largest input/output array has logarithmic size $x+y$; the depth-first implementation and the integer-ring lifting above give the space bound.

The bilinear tensor decomposition can be executed with any of its three legs designated as the output. Permuting or transposing the three legs therefore supplies every ordering of the dimension triple, with the input/output arrays still charged. Finally,

$$1 > \frac{335}{428} + 10^{-5} > \frac{50}{107} > 0.$$

The rearrangement inequality says that the largest permutation of the second support assigns these three weights to x, y, z in decreasing order; the first support has three distinct permutations. Thus the support maximum is exactly F_ε, not an uncharged choice of an arithmetic algorithm. In particular the full-array branch is retained. □

What is uniform and what remains asymptotic. The accuracy, schemes, number field, basis, and rational multiplication table are fixed before any SAT input. Their effective search can be performed once or included as a finite initialization. Matrix sizes and entry bit lengths vary; their handling is uniform and deterministic. The proof establishes an eventual complexity bound, not a supplied fast finite matrix implementation or a practical crossover estimate. In particular the enormous unknown fixed search and scheme constants must not be described as negligible in measured implementations. They are legitimate fixed constants in the asymptotic theorem.

Pinned primary sources. The source files and their blob hashes are:

1. October 2 square theorem and exact-rank/algebraic-coefficient remark: *An Upper Bound of 9/4 for the Matrix Multiplication Exponent*; blob `4492e66c707b9329f763c46e7cc113ff21d2edc0`.
2. September 24 strict dual theorem: *Complex Matrix Multiplication Below 2.258 and Rectangular Bounds*, introduction; blob `6b5dd1a6ac7f36b8dc046833960735094ca4e73f`.
3. Same paper, coefficient-equation descent and characteristic-zero dual-bound corollary: field-transfer section; blob `b7fa963d5e2659639b047e3a7dc179a090acd40a`.

5 Exact certificates for adaptive contraction plans

Fix an exact mode-weight allocation. A plan partitions the r group labels into three ordered blocks. Inside each block, recursively split after its floor midpoint and form the two boundary tables. Finish by contracting the three complete block tables. A merge has parts A, B, O, where O is the complement of $A \cup B$. Its three index classes are the unordered edges between those parts. The final operation has the three final blocks as its parts. There are $r - 2$ operations in a complete plan.

For the backend of Theorem 4.4, set

$$H_0 = 21\,400\,000.$$

The nine integer support triples are the distinct permutations of

$$(21\,400\,107, 21\,400\,107, 0), \qquad (21\,400\,000, 16\,750\,214, 10\,000\,000). \tag{6}$$

A support vector is zero on edges inside an operation part, and takes the corresponding coefficient on each of its three cross classes. Let $\mathcal{V}_j$ contain all $9(r-2)$ supports of plan j. The local operation

exponent of that plan is bounded by

$$F_j(g) = \frac{1}{H_0} \max_{v \in \mathcal{V}_j} v \cdot g.$$

This includes every internal merge. It does not count only the final triangle. The input/output branch in (6) also pays for every stored boundary table.

Lemma 5.1 (Weighted coupled-pair certificate)**.** *Let there be $q \geq 2$ plans, with fixed nonnegative integer weights $w_1, \ldots, w_q$ and $W = \sum_j w_j > 0$. For $i < j$ compute*

$$A_{ij} = \max_{v \in \mathcal{V}_i,\, w \in \mathcal{V}_j} \left(w_i^2 \|v\|_2^2 + w_j^2 \|w\|_2^2 + 2(q-1) w_i w_j (v \cdot w) \right).$$

Choosing the least-cost plan gives local time and space coefficient

$$\beta_r^2 = \frac{2 \sum_{i<j} A_{ij}}{(q-1) W^2 H_0^2 r^2} \tag{7}$$

relative to $\mathcal{K}(M, u)$. For one plan the valid squared coefficient is $2 \max_{v \in \mathcal{V}_1} \|v\|_2^2 / (H_0^2 r^2)$.

Proof. For a given $g \geq 0$, choose v_j attaining the maximum for plan j. The exact identity

$$(q-1) \left\| \sum_j w_j v_j \right\|_2^2 = \sum_{i<j} \left(w_i^2 \|v_i\|_2^2 + w_j^2 \|v_j\|_2^2 + 2(q-1) w_i w_j v_i \cdot v_j \right)$$

holds because each diagonal term is counted $q-1$ times and each cross term once with its required multiplier. Therefore the squared norm is at most $\sum A_{ij}/(q-1)$. Also

$$\min_j F_j(g) \leq \frac{1}{W H_0} \left(\sum_j w_j v_j \right) \cdot g.$$

Apply (3). Only a fixed finite number of supports and support sums is possible, so the inherited error is uniform. One need not suppose the independently maximizing support pairs are jointly consistent. An upper bound on their sum is sufficient. The single-plan case follows directly from the same entropy estimate.

To choose a plan, write $d_{\{a,b\}} = |\mathcal{I}_{ab}|\,|\mathcal{I}_{ba}|$. Compute for each plan the integer

$$P_j = \max_{v \in \mathcal{V}_j} \prod_e d_e^{v_e},$$

and choose the smallest, breaking ties by its index. Its logarithm is $H_0 F_j(g)$. The number of supports, their coefficients and denominators are fixed, so these integers have polynomial bit length. Selection introduces no comparison oracle for real logarithms. All input/output tables are covered by the first support family. Sequential matrix execution then bounds local space by local time on the exponential scale. □

The identity is a finite mathematical tool. The new work consists of the updated backend, the new fifteen-group construction, their exact certificates and the paid global schedules. Averaging and quadratic norm identities themselves are classical.

5.1 The new fifteen-group cyclic construction

Start with the ordered blocks

$$(13, 8, 12, 10, 1) \mid (2, 14, 0, 5, 6) \mid (3, 4, 7, 9, 11). \tag{8}$$

For $j = 0, \ldots, 14$, add j modulo 15 to every entry, preserving the within-block order. Use all fifteen resulting plans with equal weights. Each five-element block is built by the prescribed recursive split, so the internal order matters. There are 117 supports per plan.

Exact enumeration of all $\binom{15}{2}117^2 = 1\,437\,345$ support pairs gives

$$\boxed{\beta_{15}^2 = \frac{10\,797\,642\,389\,573\,327}{38\,640\,375\,000\,000\,000}}. \tag{9}$$

The generator, all plans, pair maxima and maximizing indices are supplied. The checker uses arbitrary-precision integers throughout. An independent implementation explicitly reconstructs the four merges inside each five-element block and confirms the same fraction. This is an exhaustive finite certificate, not a sample of entropy vectors or a numerical optimization output. The seed was found by finite exploratory search; no global optimality of the seed or schedule is claimed.

5.2 Affine nine and cyclic thirteen

The four parallel classes of the affine plane $\mathbb{F}_3^2$ give the following useful complete plans:

$$\begin{gathered}(0, 1, 2) \mid (3, 4, 5) \mid (6, 7, 8),\\ (0, 3, 6) \mid (1, 4, 7) \mid (2, 5, 8),\\ (0, 5, 7) \mid (1, 3, 8) \mid (2, 4, 6),\\ (0, 4, 8) \mid (2, 3, 7) \mid (1, 5, 6).\end{gathered}$$

These plans are inherited from the latest affine packets. The updated backend yields a new coefficient. Put $a = 335/428 + 10^{-5}$ and $b = 50/107$. The exact value is

$$\beta_9^2 = \frac{3(1 + a^2 + b^2) + 2(1 + a + b)^2}{54} = \frac{357\,623\,761\,181\,449}{1\,236\,492\,000\,000\,000}. \tag{10}$$

There is a geometric explanation for the expression: a final second-branch support has squared norm $9H_0^2(1 + a^2 + b^2)$, and supports from transverse directions have inner product $2H_0^2(1 + a + b)^2$. Every intersection of their three cross classes has exactly two edges. The largest coupled pair value is 42 914 851 341 773 880. The exact checker exhausts all $6 \cdot 63^2 = 23\,814$ pairs and checks that the internal-merge cases do not exceed that value. All four final supports attain it jointly; this certifies sharpness of this particular average-norm calculation, not optimality for actual SAT instances or all plans.

For thirteen groups use all cyclic translates of the existing ordered seed

$$(11, 2, 8, 9, 7) \mid (10, 1, 0, 5) \mid (6, 4, 3, 12).$$

With the new backend, the same exact method gives

$$\beta_{13}^2 = \frac{36\,196\,226\,703\,680\,947}{125\,767\,265\,000\,000\,000}. \tag{11}$$

There are 99 supports per plan and $78 \cdot 99^2 = 764\,478$ checked pairs.

5.3 Ten-group cache and small tails

The ten-group stage retains the nine plans and mixing weights of [4]. The weights, in plan order, are

$$(137, 124, 135, 120, 108, 119, 106, 87, 64), \qquad W = 1000.$$

The complete ordered plans are in `data/new_backend_weighted10.json`. They are included rather than read from a mutable external location. Applying Lemma 5.1 to the new supports gives

$$\beta_{10}^2 = \frac{13\,753\,456\,054\,476\,639\,694\,573}{45\,796\,000\,000\,000\,000\,000\,000}. \tag{12}$$

All $36 \cdot 72^2 = 186\,624$ pairs are checked. This does not require the optional unary-potential reparameterization explored during the review.

The small-tail plans are $(0,1,2) \mid (3,4) \mid (5,6)$, $(0,1) \mid (2,3) \mid (4,5)$, and $(0,1) \mid (2) \mid (3)$. Their coefficients are

$$\beta_7^2 = \frac{120001200003}{245000000000}, \quad \beta_6^2 = \frac{4444488889}{10000000000}, \quad \beta_4^2 = \frac{40000400001}{80000000000}.$$

The input/output branch determines these bounds. Omitting it would incorrectly shrink a cost that includes materializing the tables.

6 Fully charged recursive schedules

All size parameters in this section refer to the original instance. In particular, we do not restart a low-degree schedule at a child whose tensors were created at an earlier depth. Write

$$\lambda = \lceil \log_2 s \rceil, \quad \Lambda = \lambda - \lceil \sqrt{\lambda} \rceil, \quad b = \lceil 6 \log_2 \lambda \rceil, \quad H = \lceil s/\lambda \rceil.$$

Directly contract instances with at most H wires. For all larger calls at global depth d, choose

$$u_d = \lfloor \eta_d \Lambda \rfloor, \qquad a_d = \sum_{j=0}^{d} u_j + (d+1)b. \tag{13}$$

The finitely many sizes with nonpositive tapers are also treated directly. All these choices define one uniform algorithm after fixing the backend accuracy. Rounding the logarithms can be implemented with integer bit lengths and integer power comparisons.

Both schedules have four initial stages followed by a four-group tail. Set $q = 63/100$. If A_* denotes the total taper budget and $A_3 = \sum_{j=0}^{3} \eta_j$, then

$$r_d = 4, \qquad \eta_d = (A_* - A_3)(1-q)q^{d-4} \quad (d \geq 4). \tag{14}$$

	Generic root degree $s^{9/10}$		Low root degree $s^{1/12}$	
Stage d	r_d	η_d	r_d	η_d
0	10	301/20000	15	2149/20000
1	13	287/25000	10	2337/100000
2	10	357/100000	7	407/20000
3	7	193/50000	6	109/10000
4	4	14837/2500000	4	140341/10000000
Total A_*		1/20		1/5
Tail budget $A_* - A_3$		401/25000		3793/100000

6.1 Packing gates and a uniform depth argument

Put $A_d = \sum_{j \le d} \eta_j$ and $A_{-1} = 0$. If the original degree exponent is δ, the root has both side degrees at most s^δ up to rounding. Below the root the fresh-side degree is at most $2^{a_{d-1}+1}$; a safe retained-side degree bound is the maximum of the original degree and all earlier packet degrees. For any fixed prefix depth, the leading exponents in the numerators of (4) are therefore bounded by

	QR	Q^2C	L_pQ^2
$d = 0$	$\delta + A_0 \log_2 r_0$	$\delta + 2A_0 \log_2 r_0$	$A_0(1 + 2\log_2 r_0)$
$d > 0$	$\max(\delta, A_{d-1}) + A_d \log_2 r_d$	$A_{d-1} + 2A_d \log_2 r_d$	$A_d(1 + 2\log_2 r_d)$.

Since an active call has $M > s/\lambda$, each strict bound below one establishes its gate, including any fixed polylogarithmic factors.

An independent rational audit certifies the following upper bounds. The displayed terminating decimals are exact rational upper bounds; entries with ellipses are shown rounded for readability.

Schedule	d	QR	Q^2C	L_pQ^2
Generic	0	.9499961	.9999922	.1150422
	1	.998174265	.21139853	.22287853
	2	.9999922	.2265144	.2300844
	3	.995339304	.220778608	.224638608
Low degree	0	.5031404833…	.9229476333…	.9470643
	1	.54203404	.97661808	.99998808
	2	.555214658	.979609316	.999959316
	3	.57012095	.9890719	.9999719

For example, the audit verifies $r^v < 2^u$ exactly to prove the rational logarithm bounds $\log_2 r < u/v$. The smallest certified prefix gaps are 39/5000000 and 149/12500000, respectively. They are positive fixed margins, though their small sizes permit extremely large asymptotic thresholds. No practical crossover claim follows.

There are boundary cases in the tail, and ignoring them would be an error. In the generic case the largest leading exponent is $9/10 + 2A_* = 1$ in QR. The use of Λ instead of λ gives the sharper bound

$$\log_2(QR) \le \lambda - \tfrac{1}{10}\sqrt{\lambda} + O((\log \lambda)^2).$$

The other two numerator exponents are at most $1/4 + o(1)$. In the low-degree case, QR has exponent at most $3/5 + o(1)$, while both remaining gates have boundary exponent $5A_* = 1$ and satisfy

$$\max\{\log_2(Q^2C), \log_2(L_pQ^2)\} \le \lambda - \sqrt{\lambda} + O((\log \lambda)^2).$$

The negative square-root margins dominate division by M, the required logarithmic factors, and accumulated integer guards.

To justify the depth used in these estimates without circular reasoning, first consider hypothetical active depths up to

$$D_\lambda = \lceil \log_{20/11} \lambda \rceil + O(1).$$

On that whole range, $d = O(\log \lambda)$, so all guards total $O((\log \lambda)^2)$. The smallest tail taper is at least a constant times

$$\lambda^{1+\ln(63/100)/\ln(20/11)}, \qquad 1 + \frac{\ln(63/100)}{\ln(20/11)} > 0.$$

Positivity follows exactly from $63/100 > 11/20$. Thus every active taper tends to infinity uniformly. The already established gates now imply tail child ratio $1/2 + o(1) \le 11/20$ for all sufficiently large s. Such an active path cannot exceed D_λ, because its size would then fall below s/λ. This closes the induction. Use relative packing slack λ^{-3}; over $O(\log \lambda)$ levels its accumulation is $1 + o(1)$. Floors are uniformly negligible.

6.2 Precision and the costs outside the main exponent

Set $a_{-1} = 0$. The inherited precision satisfies $p = O(s\lambda^2 2^{a_{d-1}})$. For a retained valuation $i < u_d$,

$$D = 2^{a_d - i} \ge 2^{a_{d-1}+b+1}, \qquad \frac{p}{MD/u_d} = O(\lambda^4 2^{-b}) = o(1).$$

The exact Hermite slope is $(1+o(1))M/u_d$, and its remaining allowance has relative size $O(\lambda^2/D) = o(1)$. This proves the uniform sharp working-precision estimate $(6+o(1))MD/u_d$ used in the entropy bound. The optimal mode tilt has scale $\sqrt{u_d}/D \to 0$ uniformly, making the generating-function remainder negligible as well.

Throughout the recursion, $a_d \le A_*\lambda + O((\log \lambda)^2)$, so packet lengths are at most $s^{A_*+o(1)}$. Mode, precision, interpolation, and packet descriptions remain polynomial in s. The source extension degree $4\lceil \log_2 s \rceil$ supplies enough residue-distinct points, because $M(2L_p - 1) + 1 < s^4$ eventually. Entry queries retain their creation precision. Fresh packet evaluation costs $2^{O(L_p \log s)} = 2^{s^{A_*+o(1)}}$, original entry work costs $2^{O(s^{9/10})}$, and direct leaves cost $2^{O(s/\lambda)}$. Each is $2^{o(s/\sqrt{\lambda})}$ under both schedules. Products of polynomial factors over the depth add only $O(\lambda \log \lambda)$ to the exponent. Description-dependent work can be charged once per call; the branching bounds do not depend on $\mathcal{L}$. The dependence on the original description therefore remains one fixed polynomial factor.

6.3 Exact global coefficients and the terminal route

Let $\mu_0 = 1$, $\mu_{d+1} = 2\mu_d/r_d$, and

$$k_d = \kappa \mu_d / \sqrt{\eta_d}, \qquad P_d = \sum_{j<d} \alpha_{r_j} k_j.$$

Unrolling the paid recurrence (5), the local route at depth d has coefficient $P_d + \beta_{r_d} k_d$; its own table space has coefficient $\beta_{r_d} k_d$. The wire masses through depth four are

	0	1	2	3	4
generic	1	1/5	2/65	2/325	4/2275
low degree	1	2/15	2/75	4/525	4/1575.

In the tail, $k_{4+j} = k_4 \gamma^j$ with $\gamma = 1/(2\sqrt{q}) = 5/\sqrt{63} < 1$. Since $\beta_4 < \alpha_4/(1-\gamma)$, the largest possible tail route is its limiting terminal coefficient

$$P_4 + \frac{\alpha_4 k_4}{1-\gamma}, \qquad \alpha_4 = \sqrt{3/8}. \tag{15}$$

It pays for every recursive table entry down to the direct leaves. For finite tail depth this is an upper bound. The unspent tail mass tends to zero, so it also specifies the leading limiting coefficient.

Charged route	Generic	Low degree
Local stage 0	28.632766276536	10.336654568635
Local stage 1	28.585713591899	9.961674795027
Local stage 2	28.484720584611	10.108413259516
Local stage 3	28.520568607891	10.174645147995
First tail stage	28.493870083097	10.206500056211
Terminal route	28.632507095859	10.336723985547

The generic maximum is the root. The low-degree maximum is the terminal route, slightly larger than its root. Accordingly the exact constants are

$$\begin{aligned} C_{\mathrm{gen}} &= \kappa\beta_{10}\sqrt{20000/301}, \\ C_{\mathrm{low}} &= \kappa\left(\sum_{d=0}^{3}\frac{\alpha_{r_d}\mu_d}{\sqrt{\eta_d}} + \frac{\alpha_4\mu_4}{\sqrt{\eta_4}(1-5/\sqrt{63})}\right), \\ C_{\mathrm{low,space}} &= \kappa\beta_{15}\sqrt{20000/2149}. \end{aligned}$$

The middle expression uses the low-degree row of the parameter table. Rational interval arithmetic proves $C_{\mathrm{gen}} < 28.633$ and $C_{\mathrm{low}} < 10.337$, as well as the indicated branch dominance. The intervals use an alternating arctangent series for π, the positive atanh series for $\ln 2$, and rational square-root enclosures. No floating-point value decides a claimed inequality.

Ancestor tables are kept while a child is evaluated. Their space adds, rather than multiplies, so the exponent coefficient is the largest local table coefficient over the recursion. In both schedules the root is largest; the decreasing geometric tail and the finitely many prefix comparisons certify this. Polynomial local workspace and recursion descriptions do not alter the coefficient. This completes the proof of Theorem 1.1 relative to the explicitly imported interface.

7 A nearly wire-preserving 3-SAT encoding

The normalized contractor parameter is a wire count. An application to SAT must pay for its actual Boolean network; replacing that parameter by the number of variables without a reduction is invalid. This section gives a direct, exact normalization with a precise wire formula.

7.1 Clause and equality tensors

First delete tautological clauses, repeated literals within a clause, and duplicate clauses. An empty clause decides unsatisfiability. A formula with no remaining clauses has 2^n satisfying assignments. Rename the remaining variables by their active indices and record the number n_u of unused declared variables. These operations preserve the count on active variables, with an eventual factor 2^{n_u}.

For each remaining clause, use its Boolean truth table on its one, two, or three occurrence wires. For a variable with d occurrences, an equality tensor on the variable side is one exactly when all its ports carry the same bit. The direct incidence network has L wires, where L is the normalized literal count. It is already exact, but a variable of large occurrence may violate the root degree premise.

Theorem 7.1 (Bounded-degree equality normalization)**.** *For any integer $B \geq 3$, a normalized 3-CNF with active occurrence counts $d_v \geq 1$ has a bipartite Boolean network of maximum degree at most B and exactly*

$$M_B = L + 2\sum_v \lfloor \frac{(d_v - 3)_+}{B-2} \rfloor \leq L\left(1 + \frac{2}{B-2}\right) \tag{16}$$

wires. Its contraction is the exact number of satisfying assignments on the active variables. Its description and entry queries are polynomial-time computable. Unused variables contribute the separate factor 2^{n_u}.

Proof. Replace the equality star of variable v by a path of

$$k_v = \max\left\{1, \lceil \frac{d_v - 2}{B-2} \rceil\right\}$$

equality nodes, all on the variable side. A path with k_v such nodes has total external-port capacity

$$k_v B - 2(k_v - 1) = k_v(B-2) + 2 \geq d_v.$$

Assign its d_v occurrence ports along the path without exceeding these capacities. Subdivide every path edge by one binary equality node on the clause side. Thus the graph stays bipartite. Each subdivision contributes two internal wires, so this variable adds $2(k_v - 1)$ wires beyond its original d_v clause incidences. The identity

$$k_v - 1 = \lfloor \frac{(d_v - 3)_+}{B-2} \rfloor$$

gives the exact formula. Summing the elementary upper bound $k_v - 1 \leq d_v/(B-2)$ proves its inequality.

Because the equality nodes form a connected tree, an assignment of external occurrence bits extends to its internal wires if and only if all external bits agree. If it extends, the extension is unique. There is therefore neither an unwanted multiplicity nor an extra restriction. Summing over occurrence wires counts exactly the satisfying assignments of the active formula. Every node has degree at most B, and each entry query tests either a clause of size at most three or equality of at most B supplied bits. Construction and entry evaluation are polynomial in the network description. □

The tree size is optimal within this particular family of degree-B equality trees with binary subdivisions: any connected tree of k variable-side equality nodes has only $k(B-2)+2$ available external ports. This is a scoped combinatorial observation, not a lower bound against arbitrary SAT encodings or tensor factorizations.

7.2 Application to the low-degree contractor

Choose

$$B = \max\{3, \lceil (\log_2(L+2))^2 \rceil\}, \qquad s = M_B + 1.$$

Then $s = (1 + o(1))L$, and $B \leq s^{1/12}$ for all sufficiently large L. Small cases may be evaluated directly. Set the initial precision to $P_0 = n_a + 1$, where $n_a \leq L$ is the number of active variables, and set $\ell = 1$. The true count is at most 2^{n_a}, so its residue modulo 2^{n_a+1} uniquely recovers it. In particular $P_0 \leq s$, as required by the interface. Multiply the recovered count by 2^{n_u} to restore unused variables. Writing this binary output costs at most a polynomial in the original $n + m$.

All original tensor entries are computable in polynomial time, which is stronger than the premise of Theorem 1.1. Since

$$\frac{s}{\sqrt{\log_2 s}} = (1 + o(1))\frac{L}{\sqrt{\log_2 L}},$$

the low-degree theorem gives Theorem 1.2. Its decision answer is simply whether the exact count is nonzero. With $L \le 3m$, monotonicity of $x/\sqrt{\log_2 x}$ for sufficiently large x gives the displayed clause and bounded-density consequences.

7.3 Comparison and limits of the normalization

A trivalent equality tree has up to $3L$ wires under the same bipartite subdivision convention. Allowing the degree cap to grow slowly makes the added internal wires $o(L)$ instead of a constant multiple of L. The construction is simple, but it is useful because the low-degree contractor permits a much larger eventual degree than a constant.

This factor saving should not be presented as an additional factor-three improvement over the strongest composition of all earlier ideas. The direct-incidence/high-occurrence conditioning method already appears in [9]. In that method, condition on every variable whose original occurrence exceeds B. There are at most L/B such variables, so it uses at most $2^{L/B}$ branches. Each residual incidence network has at most L wires and degree at most B; simplify satisfied clauses and restore any now-unused variables when counting. With the same $B = \Theta(\log^2 L)$, the branching exponent is $o(L/\sqrt{\log L})$. Its low-degree schedule can use the original budget $s = L + 1$; a smaller residual need not be treated as a fresh instance with an unjustified relative degree bound.

Thus the earlier low-degree contractor already implies a fair leading cached benchmark of $13.3796490663\ldots$ per literal. Our new $10.3367239856\ldots$ coefficient beats that benchmark. The equality encoding offers a single count-preserving network and an exact wire formula, avoiding the outer branching construction.

Only original clause and equality tensors have been split. The proof does not factor arbitrary derived tensors created by the compressed dense-formula reduction. Such tensors may lack an equality-tree representation. Consequently this normalization alone does not remove the $\log\log n$ factor in the unrestricted variable-based bound.

The supplied compiler implements this construction, validates its degree, port and bipartiteness invariants, and provides brute-force semantics for small instances. The companion verifier compares exact tensor sums with direct formula counts and checks the unique-extension property separately. It is a normalization implementation, not the complete asymptotic solver.

8 What an improvement in exponent order would require

The requested goal calls for more than optimizing constants. The source methods have two distinct losses: partition-mode entropy gives a square-root logarithm in the denominator, while the dense-formula compression contributes the worst-case $\log\log n$ factor. The results above improve the fully paid coefficient but leave both mechanisms in place. This section records the most relevant structural directions and one new obstruction to an attractive shortcut.

8.1 Why faster multiplication changes the coefficient here

The common-tilt bound is proportional to $M/\sqrt{u}$, and packing permits $u = O(\log s)$. A better exact matrix exponent changes the support coefficients and therefore the multiplier of $M/\sqrt{u}$. A different fixed finite plan changes the same multiplier. Neither change alters the singularity of the mode generating function. Even an idealized quadratic square-matrix algorithm would therefore leave this particular entropy calculation on the $s/\sqrt{\log s}$ scale.

This is an accounting statement about the present proof architecture. It is not a lower bound for 3-SAT or for all possible contractors. The global schedules also show why optimizing a local matrix cost in isolation can fail: the terminal route pays for recursive child entries, and it already determines the new low-degree time coefficient.

8.2 A precise potential route beyond square-root entropy

The independent Anthropic proof describes a graded-operator route whose current partition basis has $\log \dim V_E = \Theta(E^{1/2})$ [17]. Its recurrence balances that growth against a two-direction decomposition. At the level of that accounting, replacing the growth by $\Theta(E^{\theta})$ with $\theta < 1/2$ would replace the contraction ratio by $\rho = 2^{\theta-1}$ and the logarithmic exponent by $-\log_2 \rho = 1 - \theta > 1/2$.

That substitution is not a construction. To use it one must supply, at the same time, an exact scalar identity, effective factorizations in both directions, polynomially bounded decomposition metadata, and uniformly efficient entry access at the necessary precision. A small graded space by itself does not provide these properties. Any future proposal should state these requirements as a single complete interface and pay for converting its outputs into the next recursive inputs.

8.3 Sparse mode deletion does not preserve the required locality

One tempting attempt is to retain only square-indexed, or otherwise sparse, Heisenberg modes. This reduces the count of partitions formed from the surviving weights. It also changes the current commutator, and the original single-point locality argument then fails.

Lemma 8.1 (No proper mode deletion with finite single-point locality)**.** *Let $S \subseteq \mathbb{Z}_{>0}$, put $a_0 = 0$ and $a_n = n\,\mathbf{1}_S(|n|)$ for $n \neq 0$, and consider the formal distribution*

$$C_S(z, w) = \sum_{n \in \mathbb{Z}} a_n z^{-n-1} w^{n-1}.$$

If $(z - w)^H C_S(z, w) = 0$ for some integer $H \geq 0$, then S is empty or all of $\mathbb{Z}_{>0}$.

Proof. For $H = 0$, all coefficients vanish and S is empty. For $H > 0$, the coefficient of $z^{H-m-1} w^{m-1}$ in the product is

$$\sum_{j=0}^{H} (-1)^j \binom{H}{j} a_{m-j}.$$

Its vanishing for every integer m says that the Hth finite difference of the doubly infinite sequence a_m is zero. Newton interpolation, or induction on H, gives a polynomial $P(m)$ of degree below H agreeing with a_m on all integers. If the positive complement of S is infinite, then P has infinitely many zeros and is identically zero, so S is empty. If that complement is finite, then $P(m) = m$ at infinitely many positive integers, hence $P(X) = X$ identically. No positive integer can then be omitted, and $S = \mathbb{Z}_{>0}$. □

The sparse current commutator is exactly C_S. The cited locality proof clears the commutator with a finite power of $z - w$; the lemma shows that this proof cannot retain any nonempty proper set of modes unchanged. Keeping multiples of a fixed integer may instead admit locality at several roots, with a factor such as $(z^d - w^d)^H$. That only rescales the usual partition weights: $\log p(E/d) = \Theta(\sqrt{E/d})$, retaining the square-root growth and the same doubling ratio. A different algebra or a different exact decomposition may still work. The lemma excludes only the stated shortcut; it does not exclude sparse bases by unrelated constructions.

8.4 Other tempting shortcuts and the remaining problem

The earlier packets already explain several accounting hazards. Reducing the original target to parity does not automatically remove the Hermite recovery precision or the full mode-allocation cost. Reapplying the contractor to a compact second network also requires a factorization with acceptable degrees and entry costs; merely declaring a small number of large tensors does not supply it [6, 12]. The latest pairwise packet contains a scoped generic-oracle obstruction to summing arbitrary independently supplied tables more cheaply [4]. Such statements must not be promoted to complexity lower bounds for 3-SAT's structured tables.

There are therefore two concrete, still unproved ways to surpass the present worst-case order: construct a compatible exact decomposition with entropy exponent below $1/2$, or compress dense 3-CNF instances to an interface of $O(n)$ effective size while preserving the required entry access and precision. The equality construction proves this small-size statement for direct formulas with $L = O(n)$; it does not solve the dense case. No result in this report claims that either remaining construction has been achieved.

9 Reproducibility and proof status

9.1 What the accompanying programs verify

The archive is self-contained for its finite certificate checks and its SAT normalization tests. Those programs require only the Python standard library. Their acceptance conditions use explicit exceptions, so running Python with optimization does not disable the checks.

Component	Verification performed
`verify_new_backend.py`	All 2,274,237 support-pair choices for the supplied unweighted menus; exact fractions, witnesses, affine-nine closed form, and small tails.
`verify_weighted_ten.py`	All 186,624 weighted support pairs; rejects a lowered bound, deleted input/output supports, and an altered coefficient.
`verify_schedule.py`	Exact rational intervals for both paid maxima; prefix gates; root space dominance; tail boundary and convergence checks.
`verify_packing_gates.py`	Independent rational gate, wire-mass, depth and precision audit, with integer power comparisons.
`verify_normalization.py`	Exact Boolean semantics, unique extensions, wire formula, degree and bipartiteness, unused variables, malformed inputs, and two incorrect-gadget controls.

There are 2 460 861 exhaustive pair comparisons in the two support suites together. The backend-support checker also tests 2,197 small integer dimension-log triples against the sorted formula. These samples are a regression check; the rearrangement and interpolation proofs, rather than the samples, establish the formula for all dimensions.

The normalization suite uses 171 raw formulas and constructs 684 networks. It directly evaluates 664 complete integer tensor contractions, enumerating 361,624 edge assignments, and compares with 8,376 original variable assignments. Separately, 40 equality gadgets are checked on 2,550 boundary conditions, using 310,950 boundary/internal assignments. Twenty-five larger structural cases reach 3,994 wires. All eight invalid inputs are rejected; incorrect literal signs and broken equality are detected. These finite tests accompany the general proof in Theorem 7.1; they do not substitute for it.

The new fifteen-plan certificate also has an independent reconstruction of its trees and all 1,437,345 pair comparisons. Its fixed-width integer implementation proves that the largest possible intermediate value is below 2^{63}, then recomputes maximizing witnesses with arbitrary-size integers. The main supplied checker uses arbitrary-size integers throughout.

The suites passed both ordinary execution and `python3 -O`, with identical mathematical outputs. The final archive reruns the portable entry points. Exact result receipts and negative-control evidence are included. Neither a floating-point optimizer nor an unfinished exploratory search supplies any accepted theorem or inequality.

9.2 What is proved, imported, and not implemented

The new mathematical content is the exact characteristic-zero backend bridge, the updated finite support certificates including the cyclic fifteen-plan construction, the fully charged integer schedules, the growing-degree equality normalization, and the scoped locality lemma. The source Cauchy/Hermite contraction, packing and entropy interfaces are imported with their hypotheses; the two characteristic-zero matrix exponents are primary-source theorem dependencies. This report does not independently reprove all of those sources.

The archive does not contain a full subexponential SAT solver, explicit large matrix schemes achieving the imported exponents, or a Lean formalization of the new results. The fixed algebraic-scheme search proves effectivity, but it may have an enormous fixed cost. The packing margins and root-degree comparisons are eventual asymptotic statements, not useful finite-size thresholds or benchmark predictions.

The audit of the supplied Anthropic development records a successful upstream CI build at the requested SHA and checks its theorem statement and selected interfaces. It is not a new local Lean build. In particular, its big-O machine theorem must not be cited as formal verification of the new numerical coefficients.

9.3 Archive guide

`report.tex` and `sections/` contain the complete source; `report.pdf` is the compiled report. `code/` contains the portable checks and equality-network compiler, `data/` the exact plans and weights, and `results/` the reproduced outputs. `audit/` records independent and formal-source evidence. `provenance.json` pins the reviewed repositories and incoming archives. `PROOF_STATUS.json` separates theorem dependencies, implemented components, and unresolved objectives.

Run `python3 code/run_checks.py` from any directory to regenerate the main certificates. Add `-optimized` to run the suites with Python optimization. Run `sh build_report.sh` to rebuild the PDF using a normal LaTeX installation. Detailed commands and component scope are in `README.txt`. No downloaded prior research packet is needed to execute these finite checks.

The principal remaining objective is mathematical: produce a compatible replacement that improves the logarithmic exponent order. The exact certificates here make the improved coefficient bounds reviewable while leaving that objective explicitly open.

References

[1] OpenAI Math, *The Exponential Time Hypothesis is False*, October 25, 2026. Pinned vendored preprint and sources; in particular the quantitative contraction and reduction appendices. Upstream commit `7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a`.

[2] Vladimir Reshetnikov, ProveIt, SubexponentialSAT index and trust-status statement, snapshot `8e76d515063c4b9fbd8743a846b457e3a01bc039`. The linked `Research/` directory contains the twenty-nine placed packets reviewed here. All following ProveIt links use this same snapshot.

[3] ProveIt, incoming packet directory, including `sat_affine_cauchy`, `sat_affine_contraction`, `sat_affine_geometry`, `sat_affine_plane`, and `sat_pairwise_certificates`. The complete archive names and SHA-256 hashes are in `provenance.json`.

[4] *Pairwise certificates for exact SAT contraction*, incoming packet sat_pairwise_certificates_2026-10-26.zip. Weighted nine-plan certificate, cyclic-thirteen menu, general cached coefficient 29.81160894538347, and low-degree polynomial-space bound.

[5] *Adaptive bounds*, sat_adaptive_bounds packet. In particular `report/cached.tex`, `report/flags.tex`, and `report/interface.tex`; prior low-degree cached and general polynomial-space frontier.

[6] *3-SAT precision refinement*, report source. Sharp recovery precision and the structured-summation interface.

[7] *Asymmetric packet balancing*, report source. Near-unit packing, asymmetric degree gates, and additive precision guards.

[8] *Joint Cauchy modes*, report source. Full-block integral modes and the common-tilt entropy estimate.

[9] *Balanced packing*, report source. Whole-vertex labeling and the direct-incidence/high-occurrence conditioning application, including the closing SAT corollary.

[10] *SAT density improvement*, report source. Density-dependent contraction size.

[11] *ETH density refinement*, report source. Overlapping density-sensitive envelope and dense fallback.

[12] *3-SAT asymptotic analysis*, packet; and *Rectangular contraction*, packet. Scoped precision, enumeration, degree, and factorization barriers.

[13] OpenAI Math, *An Upper Bound of* 9/4 *for the Matrix Multiplication Exponent*, October 2, 2026. Pinned primary source. Exact-rank exponent and finite algebraic-coefficient schemes. Git blob `4492e66c707b9329f763c46e7cc113ff21d2edc0`.

[14] OpenAI Math, *Complex Matrix Multiplication Below 2.258 and Rectangular Bounds*, September 24, 2026. Main theorem statements and characteristic-zero field transfer. The strict dual exponent exceeds 93/200. Git blobs `6b5dd1a6ac7f36b8dc046833960735094ca4e73f` and `b7fa963d5e2659639b047e3a7dc179a090acd40a`.

[15] Anthropic formal-math, eth/MainResults.lean, PR 31 head `1a6eee293b9e00cb432eef6018a21f523623a9af`. Uniform finite-alphabet machine statement and exact counting bound.

[16] Anthropic formal-math, eth/Solution.lean and CountedRoute/Main.lean, same pinned head. Closed theorem and counted-recursion endpoint.

[17] *A* $2^{o(n)}$ *time algorithm for 3-SAT*, paper at the requested formalization head. Appendices A, B, D, E and H: graded decomposition, density reduction, machine accounting, and prospective entropy improvements.

[18] Anthropic formal-math, GitHub Actions verification job 121979112896 in run 40440389861. Successful build and Palomar Comparator steps for the requested head. The retrieved metadata is retained in `audit/`.